

SentinEL Technical Report

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1. Research Objective

This project investigates the adversarial robustness of machine learning-based phishing URL

2. Dataset

* Source: PhishTank (phishing URLs) and curated legitimate URL corpus

3. Baseline Model Performance

* Algorithm: Random Forest Classifier (100 estimators, max_depth=None)

4. Adversarial Attack Methodology

* Attack type: Homoglyph substitution (Unicode confusable characters)

5. Adversarial Evaluation Results

* Attacked accuracy: 81.4% (degradation: 15.8 percentage points)

6. Explainability Benchmarks

****Finding:**** Gini impurity attribution meets SOC real-time requirements (less than 2ms)

7. Limitations

* Demo dataset (14 rows) cannot reproduce full research results

8. Reproduction

Run ``python experiments/baseline_eval.py`` to train the baseline model.